QUASI-COMPACT OBJECTS AND FINITE SUBCOVERINGS OF A SURJECTION OF SHEAVES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Coproducts of sheaves as sheafified coproducts. Same chapter, Lemma lemma-colimit-sheaves. Describes the sections of a coproduct of sheaves through a covering, which is used to replace a section by one coming from a single summand.
- Quasi-compact objects of a topos. Same chapter, the definition of a quasi-compact object and its characterization by coverings. Supplies the property of a sieve or covering used in both directions.

Source reference. Original lemma 00WI: sources/stacks-sites/pinned-original-source.tex, source lines 3541–3541. Navigation pointer invoked at proof line 3541; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00WT: sources/stacks-sites/pinned-original-source.tex, source lines 3556–3556. Navigation pointer invoked at proof line 3556; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let U be an object of $\mathcal{C}$. The following are equivalent*

(1) *U is quasi-compact, and*

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(2) *for every surjection of sheaves* $\coprod_{i\in I} \mathcal{F}_i \to h_U^\#$ *there is a finite subset* $J \subset I$ *such that* $\coprod_{i\in J} \mathcal{F}_i \to h_U^\#$ *is surjective.*

Proof. Assume (1) and let $\coprod_{i\in I} \mathcal{F}_i \to h_U^\#$ be a surjection. Then id_U is a section of $h_U^\#$ over U. Hence there exists a covering $\{U_a \to U\}_{a\in A}$ and for each $a \in A$ a section s_a of $\coprod_{i\in I} \mathcal{F}_i$ over U_a mapping to id_U. By the construction of coproducts as sheafification of coproducts of presheaves (Lemma 00WI), for each a there exists a covering $\{U_{ab} \to U_a\}_{b\in B_a}$ and for all $b \in B_a$ an $\iota(b) \in I$ and a section s_b of $\mathcal{F}_{\iota(b)}$ over U_{ab} mapping to $\mathrm{id}_U|_{U_{ab}}$. Thus after replacing the covering $\{U_a \to U\}_{a\in A}$ by $\{U_{ab} \to U\}_{a\in A, b\in B_a}$ we may assume we have a map $\iota : A \to I$ and for each $a \in A$ a section s_a of $\mathcal{F}_{\iota(a)}$ over U_a mapping to id_U. Since U is quasi-compact, there is a covering $\{V_c \to U\}_{c\in C}$, a map $\alpha : C \to A$ with finite image, and $V_c \to U_{\alpha(c)}$ over U. Then we see that $J = \mathrm{Im}(\iota \circ \alpha) \subset I$ works because $\coprod_{c\in C} h_{V_c}^\# \to h_U^\#$ is surjective (Lemma 00WT) and factors through $\coprod_{i\in J} \mathcal{F}_i \to h_U^\#$. (Here we use that the composition $h_{V_c}^\# \to h_{U_{\alpha(c)}} \xrightarrow{s_{\alpha(c)}} \mathcal{F}_{\iota(\alpha(c))} \to h_U^\#$ is the map $h_{V_c}^\# \to h_U^\#$ coming from the morphism $V_c \to U$ because $s_{\alpha(c)}$ maps to $\mathrm{id}_U|_{U_{\alpha(c)}}$.)

Assume (2). Let $\{U_i \to U\}_{i\in I}$ be a covering. By Lemma 00WT we see that $\coprod_{i\in I} h_{U_i}^\# \to h_U^\#$ is surjective. Thus we find a finite subset $J \subset I$ such that $\coprod_{j\in J} h_{U_j}^\# \to h_U^\#$ is surjective. Then arguing as above we find a covering $\{V_c \to U\}_{c\in C}$ of U in $\mathcal{C}$ and a map $\iota : C \to J$ such that id_U lifts to a section of s_c of $h_{U_{\iota(c)}}^\#$ over V_c. Refining the covering even further we may assume $s_c \in h_{U_{\iota(c)}}(V_c)$ mapping to id_U. Then $s_c : V_c \to U_{\iota(c)}$ is a morphism over U and we conclude. □